

Apply filters to SQL queries

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Project description

In this project, I act as a security analyst investigating potential security incidents and identifying employee machines that require system updates. To accomplish this, I use SQL queries with filtering clauses (**WHERE**), pattern matching (**LIKE** and **%**), and logical operators (**AND**, **OR**, and **NOT**) to extract targeted records from organizational databases. Through these filtered queries, I analyze login activity for suspicious behavior and compile accurate lists of employee computers across specific departments and office locations.

Retrieve after hours failed login attempts

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = FALSE;
```

To investigate the potential security incident after business hours, I queried the `log_in_attempts` table to retrieve all records of failed login attempts occurring after 18:00. I started with `SELECT * FROM log_in_attempts` to return all columns from the table, and then applied a `WHERE` clause with the `AND` operator to ensure both conditions were met simultaneously:

- `login_time > '18:00'` filters the results to include only login attempts that took place after 6:00 PM.
- `success = FALSE` (or `success = 0`) filters the records to show only unsuccessful login attempts.

Retrieve login attempts on specific dates

```
SELECT *  
FROM log_in_attempts
```

```
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

To investigate the suspicious event on May 9, 2022, I queried the `log_in_attempts` table to review all login activity from that day and the day prior. I used `SELECT * FROM log_in_attempts` to select all columns from the table, and then applied a `WHERE` clause with the `OR` operator so that records matching either date are returned:

- `login_date = '2022-05-09'` retrieves all login attempts from the day of the suspicious event.
- `login_date = '2022-05-08'` retrieves all login attempts from the day before the event.
- Combining these two conditions with `OR` ensures that a login attempt is included in the results if it occurred on either of those two dates.

Retrieve login attempts outside of Mexico

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'MEX%';
```

To investigate suspicious login activity originating outside of Mexico, I queried the `log_in_attempts` table and filtered out all records associated with Mexico. I started with `SELECT * FROM log_in_attempts` to retrieve all columns from the table, and then applied a `WHERE` clause combining the `NOT` and `LIKE` operators with the `%` wildcard:

- `LIKE 'MEX%'` uses the `%` wildcard to match any string in the `country` column that begins with the letters "MEX", which accounts for both `MEX` and `MEXICO`.
- Adding `NOT (WHERE NOT country LIKE 'MEX%')` inverts the condition so the query returns only login attempts where the country does not start with `MEX`.

Retrieve employees in Marketing

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

To identify the employee machines in the Marketing department that require security updates, I queried the `employees` table for all Marketing staff located in the East building. I started with `SELECT * FROM employees` to retrieve all columns from the table, and then used a `WHERE` clause with the `AND` operator to filter records that meet both conditions simultaneously:

- `department = 'Marketing'` filters the records to include only employees who work in the Marketing department.
- `office LIKE 'East%'` uses the `LIKE` operator and the `%` wildcard to match any value in the `office` column that starts with "East" (such as `East-170` or `East-320`), ensuring all offices in the East building are included.

Retrieve employees in Finance or Sales

```
SELECT *  
FROM employees  
WHERE department = 'Sales' OR department = 'Finance';
```

To identify the machines requiring the security update for staff in the Sales and Finance departments, I queried the `employees` table to retrieve all records belonging to either department. I started with `SELECT * FROM employees` to return all columns from the table, and then applied a `WHERE` clause using the `OR` operator so that employees matching either condition are included:

- `department = 'Sales'` filters for employees working in the Sales department.
- `department = 'Finance'` filters for employees working in the Finance department.
- Connecting these two conditions with `OR` ensures the query returns all employees who belong to either Sales or Finance (whereas using `AND` would return no results, since a single row cannot have both values in the `department` column at the same time).

Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

To identify all employee machines that still require the final security update, I queried the `employees` table to retrieve records for everyone outside of the Information Technology department. I started with `SELECT * FROM employees` to return all columns from the table, and then applied a `WHERE` clause with the `NOT` operator to exclude IT staff:

- `department = 'Information Technology'` identifies employees who work in the Information Technology department.
- Placing `NOT` before the condition (`WHERE NOT department = 'Information Technology'`) inverts the filter so the query returns only employees whose department is anything other than Information Technology.

Summary

Throughout this activity, I used SQL queries with `WHERE` clauses, pattern matching (`LIKE` and `%`), and logical operators (`AND`, `OR`, and `NOT`) to filter database tables for security investigations and system updates. By querying the `log_in_attempts` table, I isolated after-hours failed logins, login activity on specific dates, and login attempts originating outside of Mexico. By querying the `employees` table, I accurately identified employee machines across specific departments and buildings—including Marketing, Finance, Sales, and all departments outside of Information Technology—to ensure required security updates were properly targeted.